Human-like Artificial Agency Should Incorporate Curiosity, Compression, and Resource Constraints

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Abstract

Current AI systems (agents) increasingly reason, use tools, search for information, communicate, and act over multiple steps, yet many benchmarks emphasize final task success while treating observation, deliberation, communication, memory, and action costs as secondary. This position paper argues that in order to develop more capable and human-like artificial agents, the constraints on observation, action, deliberation, and communication under resource constraints should be incorporated into a system driven by curiosity-as-learning-progress. We propose a formalization of *Artificial Agency* for studying agents through predictive compression, curiosity-as-learning-progress, control, and interface quality under resource budgets. We provide performance–cost and interface-quality metrics and outline falsifiable experiments comparing adaptive compute policies, latent recurrence, and self-communication.

1 Introduction

Biological agency develops under reality embedded interaction, finite compute and memory, constrained sensing and actuation, and the continual need to act under uncertainty [Brooks, 1991, Wilson, 2002, Pfeifer and Bongard, 2006, Kaelbling et al., 1998, Friston, 2010, Lake et al., 2017, LeCun, 2022]. We propose that AI should be similarly designed as part of a larger human–tool system that improves sensing, understanding, and actuation under realistic constraints, while remaining efficient, interpretable, and aligned.

Current ML systems increasingly spend variable test-time compute, use tools, request observations, produce private reasoning traces, and act in interactive environments [Wei et al., 2022, Nye et al., 2021, Nakano et al., 2021, Yao et al., 2023b, Schick et al., 2023, Snell et al., 2025]. Yet benchmarks often report final task score while treating observation, communication, memory, and deliberation costs as implementation details [Hendrycks et al., 2021, Srivastava et al., 2023, Jimenez et al., 2024, Liang et al., 2023]. This paper argues that the constraints these pose and their associated compute cost should become primary for the design and evaluation of *artificial agents*.

This framing emphasizes *agency gain* and *interface friction* simultaneously. Increasing raw capability alone is not sufficient if the resulting system is hard to steer, poorly coupled to human intent, or unable to operate efficiently under the time, energy, and communication budgets in real environments. We treat agency as a property of the coupled system [Clark and Chalmers, 1998, Lieder and Griffiths, 2020, Amershi et al., 2019, Shneiderman, 2020, Hadfield-Menell et al., 2016], as many useful AI deployments require increasing the effective agency of a coupled human–AI system.

There is a deep connection between the "it-from-bit" view of physics which ties observer and environment through yes/no questions [Wheeler, 1990] and the curiosity objective presented in Schmidhuber [2010]—increasing the rate of compression by seeking patterns (observations) on which prediction improvement is maximal given current capabilities. This avoids completely random

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

patterns or patterns which are easy to compress and is analogous to the rate at which we can ask-and-answer questions about the universe. Our position builds on top of this core objective as the basis of intelligent agents.

Position: **The AI community should evaluate and design AI agents as resource-constrained curiosity-as-prediction-progress driven interactive systems that must decide when to observe, act, deliberate, and communicate.**

Our contributions are:

1. We argue that AI systems and agents should be designed and evaluated as resource-constrained interaction under a curiosity-driven objective.
2. We formalize observation, action, deliberation, communication, and memory as resources in an agent-environment setup.
3. We propose performance–cost and interface-quality metrics for comparing agents.
4. We outline an experimental agenda using observe/act/deliberate environments.

2 Human-like Artificial Agency

2.1 Human frames of reference

Many failure modes of current systems are difficult for people to anticipate precisely because the systems are trained and deployed under non-human conditions. Internet-scale next-token training with effectively superhuman memory for textual regularities and weak grounding in action can produce competent behavior, but it does not reproduce the developmental constraints that shaped human cognition. Human intelligence is structured by limited memory, limited sensing and actuation bandwidth, limited processing speed, partial observability, and the need to rely on tools and other people for increasing agency and cognitive offloading [Miller, 1956, Baddeley, 1992, Clark and Chalmers, 1998, Kirsh and Maglio, 1994, Risko and Gilbert, 2016, Ryan and Deci, 2000, Lake et al., 2017]. Therefore parts (or dimensions) of human cognition must necessarily arise precisely due to the restrictions imposed by our specific biological instantiation from a more general intelligence manifold.

This suggests a distinction between *capability* and *distance from human constraints*. Two systems may achieve similar performance (intelligence) while occupying different regions of a constraint manifold defined by sensing modalities, action affordances, energetic pressure, temporal scale, and memory budget. We propose measuring both overall competence (e.g., predictive performance, empowerment, control) and constraint proximity, because these jointly affect interpretability, collaboration quality, and failure modes [Legg and Hutter, 2007, Hernández-Orallo, 2017, Chollet, 2019, Lieder and Griffiths, 2020]. By representing intelligence as such a manifold, one may quantify overall competence as the magnitude of a point on the manifold, providing a scalar comparison between human and artificial intelligence. Human-likeness (or alienness) can be measured with a distance metric between human and AI inhabited regions of the manifold.

The reason it is hard to encode into AI the fundamental aspects of human intelligence is that they are part of a fundamentally human frame of reference. What is curious or novel to us is defined by our scale in time and space, by evolutionary programming, by biological considerations, by the properties of our sensing and actuating mechanisms; indeed even by our level of intelligence or cognition—testament to this is the fact that looking at the arch of human (and animal) history, curiosity and inquisitiveness has always shifted—adjusted—depending on our cognitive abilities [Loewenstein, 1994, Gottlieb et al., 2013]. A human 1000 years ago could hardly be curious about differential topology. An agent can only be curious with regards to what it can sense, understand and act upon—where all three include tools forming part of the extended self. We posit that it is incomplete to evaluate intelligence without specifying the constraints under which it operates. We expect this framework to guide experiments on interfaces (between agent and environment), collective cognition, alignment, and new forms of intelligences.

2.2 Alignment

We connect this formal lens to the notion of alignment and sparsity as one phenomenon across scales. We loosely define sparsity as an information-transfer bottleneck between parts of a system [Tishby et al., 1999], and define alignment as the degree of influence and mutual predictability through that bottleneck. The level of sparsity fundamentally limits alignment, but alignment is also affected by distance on the intelligence manifold, i.e. the farther apart two systems are the harder for them to align. This view applies to cells in the human body, layers in a neural network, brain regions, human groups and institutions, and human-AI systems, giving us one common framework for studying coupled systems across scales [Amodei et al., 2016, Hadfield-Menell et al., 2016]. A more rigorous account of alignment should be grounded in communication structure (sparsity) and shaped by the intelligence manifold.

As agency / intelligence increases, alignment becomes more strongly bidirectional, because any system with enough power to shape another through a shared interface enters a regime of reciprocal influence through that same interface [Hadfield-Menell et al., 2016, Amershi et al., 2019, Shneiderman, 2020]. Influence and alignment therefore co-produce one another. For example, a human who intends to influence (align) another human will often find it itself needs a similar degree of alignment — this view already is and will be increasingly important in a world teeming with superhuman AI. We urge researchers to formally study the intelligence / agency manifold so that we can model alignment more effectively and design systems closer to us.

2.3 Curiosity, compression, and progressive capability expansion

We adopt a learning-progress view of curiosity: intrinsically valuable patterns are neither random nor trivial, but those for which the agent can currently improve compression or prediction [Schmidhuber, 2010, Oudeyer et al., 2007, Gottlieb et al., 2013]. This makes curiosity inherently capability-relative. An agent can only be curious about patterns it can (in some sense) sense, model, or act upon, and the difficulty of useful tasks should grow with the agent's capabilities rather than being fixed in advance. This view aligns with developmental perspectives in which exploration and play are active processes of skill acquisition rather than passive consumption of novelty [Ryan and Deci, 2000, Oudeyer et al., 2007].

On this view, curiosity creates pressure for *capability expansion*. If learning progress is limited by sensing, actuation, memory, or compute, the agent should prefer interventions that relax whichever bottleneck most improves future learning progress, subject to cost. This links curiosity to empowerment, control, and communication: an agent may benefit from seeing more, doing more, delegating more, or coordinating with others and tools more effectively. Because the world presents patterns over space and time, this pressure naturally favors longer-horizon prediction and increasingly structured internal models [Ha and Schmidhuber, 2018, Hafner et al., 2023]. Purely static benchmarks miss constraints that become central in deployed interactive systems such as latency, sensing, actuation, energy, and communication.

An agent that is aware of its limitations may direct its curiosity-seeking to discovering what happens when removing some of those limitations, since this is a future fundamentally harder to predict than anything within current interfacing limitations. This is a sort of meta-objective: the agent not only wants to predict the world as it can interact with it at any point, but it should want to predict the kinds of questions it may ask about the world when its observation and actuation capabilities increase.

2.4 Energy–performance frontiers

Many AI systems become practically useful only when they solve tasks within meaningful time and energy budgets. This motivates explicit energy/compute–performance frontiers for how much task performance is obtained per unit cost [Rissanen, 1978, Grünwald, 2007, Hutter, 2005, Patterson et al., 2021]. We emphasize adaptive allocation of computation and communication. In many tasks, the optimal policy is not to spend maximal compute at every step, but to modulate effort based on uncertainty, expected value of information, and opportunity cost [Graves, 2016, Banino et al., 2021, Callaway et al., 2018, Lieder and Griffiths, 2020]. This efficiency emphasis also motivates the later focus on private deliberation tokens, action tokens, and observation tokens as interchangeable budgeted resources.

3 Formal setup and proposed metrics

We model an embedded agent interacting with an environment as a partially observed controlled process [Smallwood and Sondik, 1973, Kaelbling et al., 1998]. Let X_t denote the (generally hidden) environment state, S_t the agent internal state (memory + policy state (θ_t) + latent world-model state), O_t the observation, and $A_t := (U_t, V_t)$ the action.

$$
\begin{aligned}
O_t &\sim p_O(\cdot \mid X_t;\, c_t^O), && (1)\\
S_t &\sim p_S(\cdot \mid S_{t-1}, O_t, U_{t-1}, V_{t-1};\, c_t^C), && (2)\\
(U_t, V_t) &\sim \pi(\cdot \mid S_t;\, c_t^A), && (3)\\
X_{t+1} &\sim p_X(\cdot \mid X_t, U_t), && (4)
\end{aligned}
$$

where c_t^O, c_t^A, c_t^C denote observation, actuation, and compute/interface capacity constraints (e.g., sensor bandwidth, action alphabet/determinism, context length, update budget). These capacities themselves may be modeled as dynamic and agent-controllable.

$$
c_{t+1}^O \sim p_{c^O}(\cdot \mid c_t^O, V_t), \qquad c_{t+1}^A \sim p_{c^A}(\cdot \mid c_t^A, V_t), \qquad c_{t+1}^C \sim p_{c^C}(\cdot \mid c_t^C, V_t). \tag{5}
$$

We view p_S as the agent-side update operator and allow it to include both recurrent/memory updates and (optionally) learning updates of parameters θ_t (e.g., a bounded number of gradient/Bayes-style update steps), all priced under the compute/interface budget c^C.

3.1 Learning-progress intrinsic reward

We adopt a Schmidhuber-style learning-progress signal [Schmidhuber, 2010], and evaluate progress on future observations that were not used for the update. Let H denote the update-window length and K the evaluation horizon. For any integers $a < b$, define the interaction segment

$$
\mathcal{D}_{a:b} \;:=\; \big(O_a, A_a, O_{a+1}, A_{a+1}, \ldots, O_{b-1}, A_{b-1}, O_b\big), \tag{6}
$$

where $A_t = (U_t, V_t)$ as in the formal setup.

Let p_θ denote the agent's predictive model for observations, which maps a history to a distribution over the next observation:

$$
p_\theta(O_{t+1} \mid \mathcal{D}_{a:t}) \;\equiv\; p_\theta(O_{t+1} \mid O_{a:t}, A_{a:t})\,. \tag{7}
$$

Let θ_t^- denote the model before an update at time t, and let

$$
\theta_t^+ \;:=\; \mathcal{U}\big(\theta_t^-; \mathcal{D}_{t-H:t}\big) \tag{8}
$$

denote the model after applying a bounded update operator $\mathcal{U}$ to the recent interaction segment $\mathcal{D}_{t-H:t}$. The update operator may correspond to gradient steps, Bayesian updating, recurrent-state adaptation, memory writing, or any other agent-side update, with its cost accounted for in the main resource objective.

We instantiate the horizon-K predictive loss as the cumulative negative log-likelihood on future observations:

$$
\mathcal{L}_{\text{pred}}(\theta; \mathcal{D}_{t:t+K}) \;:=\; \sum_{k=1}^{K} \ell\big(p_\theta(\cdot \mid \mathcal{D}_{1:t+k-1}),\, O_{t+k}\big)\,, \qquad \ell(p, o) := -\log p(o). \tag{9}
$$

The learning-progress intrinsic reward is then

$$
r_t \;:=\; \mathcal{L}_{\text{pred}}(\theta_t^-; \mathcal{D}_{t:t+K}) \;-\; \mathcal{L}_{\text{pred}}(\theta_t^+; \mathcal{D}_{t:t+K}), \qquad t > H. \tag{10}
$$

For $t \le H$, we set $r_t := 0$. In online settings, this reward is assigned after the future segment $\mathcal{D}_{t:t+K}$ has been observed.

Crucially, the update segment $\mathcal{D}_{t-H:t}$ and the evaluation targets $O_{t+1:t+K}$ are disjoint. Thus the reward measures improvement in predictive compression on future observations. This rewards *improvement* in predictive compression rather than complexity or surprise alone [Schmidhuber, 2010, Ha and Schmidhuber, 2018, Hafner et al., 2023].

The final objective combines intrinsic reward with costs:

$$J(\pi, p_S) \;=\; \mathbb{E}\Bigg[\sum_{t=1}^{T} \gamma^{t-1}\Big(r_t - \lambda_O C_O(t) - \lambda_E C_E(t) - \lambda_C C_C(t) - \lambda_M C_M(t)\Big)\Bigg]. \tag{11}$$

Where:

- $C_O(t)$: observation-processing cost (e.g., observation tokens/bits ingested at time t),
- $C_E(t)$: actuation/maintenance cost (e.g., energy penalty for executing U_t),
- $C_C(t)$: compute/deliberation/learning cost (e.g., FLOPs, number of deliberation tokens, or number of update steps executed at time t)
- $C_M(t)$: a memory-maintenance cost motivated by finite-state maintenance and thermodynamic viewpoints on information processing [Landauer, 1961, Bennett, 1982, Still et al., 2012, Parr et al., 2022].

3.2 Control, empowerment, and plasticity

We relate predictive compression to control and empowerment. A standard k-step empowerment definition is the channel capacity from action sequences to future observations [Klyubin et al., 2005a,b, Salge et al., 2014]:

$$\mathcal{E}_k(s_t) \;=\; \max_{p(a_{t:t+k-1})} \mathrm{I}\big(A_{t:t+k-1}; O_{t+k} \,\big|\, S_t = s_t\big). \tag{12}$$

Plasticity (observation-to-action influence) has been similarly formulated in prior work using directed information [Abel et al., 2025]. We emphasize that plasticity is environment pressure on the agent.

3.3 Unification as interface quality

We introduce *unification* as reduction of sensing/acting bottlenecks between agent and environment. We formalize unification as an interface-quality concept [Tishby et al., 1999].

Let b_O, b_A denote observation and action bottleneck parameters (coarsening, noise, latency, cardinality constraints, etc.). These bottleneck parameters are simply an alternative parameterization of the capacity constraints defined earlier: b_O indexes the observation channel family $p_O(\cdot \mid X; b_O)$ with effective capacity $c^O(b_O)$, and b_A indexes the actuation/policy interface with effective capacity $c^A(b_A)$. We likewise let b_L parameterize a communication/self-communication interface whose cost is accounted for under the same budgeting terms already present in equation (11). Define a task-relative *unification score*

$$\mathcal{U}_t \;:=\; w_O \underbrace{\left(1 - \frac{\mathcal{R}_O(b_O)}{\mathcal{R}_O^{\max}}\right)}_{\text{observation losslessness proxy}} + w_A \underbrace{\left(1 - \frac{\mathcal{R}_A(b_A)}{\mathcal{R}_A^{\max}}\right)}_{\text{action authority proxy}} + w_L \underbrace{\left(1 - \frac{\mathcal{R}_L(b_L)}{\mathcal{R}_L^{\max}}\right)}_{\text{communication bottleneck proxy}}, \tag{13}$$

$$w_O + w_A + w_L = 1, \qquad \mathcal{R}_{\cdot} \geq 0 \tag{14}$$

where $\mathcal{R}_{\cdot}$ are calibrated inefficiency measures induced by each bottleneck. The exact choice of $\mathcal{R}$ is environment-specific. This way we can test whether improved agents systematically spend resources to increase effective interface quality when allowed to do so.

One way to define the observation-interface inefficiency is equivocation:

$$\mathcal{R}_O \;:=\; H(X|O), \qquad \mathcal{R}_O^{\max} = H(X). \tag{15}$$

$\mathcal{R}_A$ and $\mathcal{R}_L$ can be similarly defined.

3.4 Language as an information bottleneck

We view language as a selective, lossy, resource-constrained communication channel whose value depends on the environment and on the presence of other agents/tools. This perspective is consistent with extended cognition [Clark and Chalmers, 1998], cognitive offloading [Kirsh and Maglio, 1994, Risko and Gilbert, 2016], communication under resource rationality [Lieder and Griffiths, 2020], and recent reasoning systems that separate or modulate explicit verbal traces from latent computation [Nye et al., 2021, Wei et al., 2022, Graves, 2016, Banino et al., 2021, Wang et al., 2026]. It also helps separate two questions that are often conflated in language-model practice: whether language is useful as a training substrate for acquiring broad abstractions, and whether language is the best *online* medium for every intermediate computation once an agent is acting in a world.

From the perspective of a single embedded agent, language is one possible compression channel among many. Its value increases sharply when the environment contains other agents, humans, and tools whose behavior can be altered through communication. In such settings, communication is simply another action class whose utility depends on whether the expected future gain in prediction, control, or coordination exceeds its cost in time, bandwidth, and energy. We therefore treat (self-)communication as an action set among many, not a universally privileged mode.

Language-like communication can also be useful internally. A compact, lossy self-directed channel may support deliberation by forcing the system to encode intermediate structure in a form that is easier to re-read, monitor, and compose [Nye et al., 2021, Wei et al., 2022, Zelikman et al., 2022]. One plausible mechanism giving rise to such self-communication is a communicator–interpreter loop: to produce useful messages (even to itself), the agent must model how those messages will be interpreted, which can induce a partial third-person view of its own intermediate states. This motivates concrete experiments comparing latent-only recurrence with explicit private token channels (i.e. standard verbal reasoning traces).

3.5 Flexible deliberation

All agentic AI (i.e., LLM, VLM, VLA, etc.) systems should have reasoning modules after having "learned" language through next-token prediction [Brown et al., 2020, Wei et al., 2022, Nye et al., 2021, Snell et al., 2025]. The purpose of a reasoning module however is not to produce a stream-of-words, but rather to allow for deliberation without action. Therefore when a model is post-trained for reasoning on various tasks (including language itself) it should be able to flexibly produce sequential hidden state updates without the use of language tokens. Such a model would have the flexibility to choose when to produce and when to avoid self-communication through language and do pure deliberation instead (i.e., not through tokenization bottlenecks). Deliberation should also be expanded to any token modality, including vision and action tokens. This is analogous to visual or motor imagery in neuroscience [Jeannerod, 1994, Kosslyn et al., 2001].

Such systems might observe various types of input tokens and be able to produce and interleave these at any step. During production there should not be any fixed order, similar to how a human might pause mid-sentence to do a tiny deliberation loop (with or without explicit inner verbalization). Allowing the model to flexibly start "thinking" while it has not finished ingesting a paragraph, or continue thinking while producing an output could help with error recovery and self-monitoring.

In the most general case we propose a modality-agnostic token taxonomy with three roles per modality $m \in \{V, S, T, A, \dots\}$:

$$m_i \text{ (input)}, \qquad m_s \text{ (private/self)}, \qquad m_o \text{ (public/output)}.$$

This yields a unified interface where the policy can interleave incoming observations, external actions, and internal deliberation:

$$\underbrace{V_i, S_i, T_i}_{\text{incoming}} \;\Leftrightarrow\; \underbrace{H, V_s, S_s, T_s, A_s}_{\text{deliberation / imagery / reasoning}} \;\Leftrightarrow\; \underbrace{V_o, S_o, T_o, A_o}_{\text{communication / output}},$$

where V denotes vision, S denotes audio, T denotes text, and A denotes action tokens. H refers to a continuous output hidden state. In the ideal case the policy should learn *when private tokens are worth emitting* under budget, and *which modality is most effective* for deliberation at any given step.

This concretely supports our compute-efficiency criterion. It is an open and difficult question how to train such a flexible policy given LLM paradigms.

3.6 Efficiency and pareto optimality

A useful diagnostic for artificial agency is an energy/compute–performance frontier [Strubell et al., 2019, Snell et al., 2025]. Let P be task performance and let C denote a normalized cost. In practice, C may include several components, such as training cost, inference-time compute, memory/storage cost, and human-interaction cost. We can summarize each policy by (C, P) and compare it to an empirical Pareto frontier $\mathcal{F}$. When a scalar summary is useful, one option is the normalized distance to the frontier:

$$d_{\mathcal{F}}(C, P) \;=\; \inf_{(c,p)\in\mathcal{F}} \left\|(\tilde{C}, \tilde{P}) - (\tilde{c}, \tilde{p})\right\|_2, \tag{16}$$

where tildes denote task-specific normalization of cost and performance.

We propose comparing systems by their Pareto frontiers: a stronger agent should achieve better performance at comparable cost, lower cost at comparable performance, or better adaptive movement along the frontier as task difficulty varies.

4 Hypotheses

Here we outline 5 central hypotheses that our framework makes and which then can be tested using the experimental agenda in Section 5.

4.1 H1: Alignment of objectives

In resource-bounded agents acting in a partially observable environment, increasing learning progress on future observations also tends to increase useful control over environmental degrees of freedom, and vice versa. Moreover, optimizing equation (11) with respect to agent-side observations implies similar learning progress (or control) over the true environment hidden state from which observations are drawn, when interfaces and bottlenecks are flexible enough to allow for losslessness (i.e. invertibility) in the limit.

4.2 H2: Boundary pressure toward unification

When an agent can invest resources to modify its sensing/acting/communication interfaces, optimization of equation (11) will allocate resources toward interface improvements that increase long-horizon learning progress and control, yielding monotonic gains in unification $\mathcal{U}$ until costs dominate. In an idealized limit where (i) environment complexity is finite and (ii) marginal interface-improvement costs scale favorably, the optimizer can drive $\mathcal{U} \to 1$ (maximal effective interface quality).

4.3 H3: Constraint-induced predictive/control pressure

Under continued viability constraints (cannot cease interaction with environment), coarse/noisy interfaces, nonzero costs for action, memory maintenance, and frequent policy updates, agents are driven toward better prediction and selective control because these reduce costly adaptation (plasticity) and wasted computation. Therefore even without directly providing the intrinsic reward of equation (10), an agent coupled to a realistic environment that has increasing entropy will nevertheless be forced to optimize toward it in order to maintain its (low-entropy) state. This hypothesis is motivated by information-theoretic and thermodynamic arguments linking predictive state representations to energetic efficiency [Still et al., 2012, Friston, 2010, Parr et al., 2022, Ortega and Braun, 2013].

4.4 H4: Adaptive compute optimality

We predict that a meta-controller that dynamically allocates observation, action, and deliberation compute should outperform fixed schedules under the same total budget, especially when task difficulty and observability vary across timesteps.

4.5 H5: Self-communication bottleneck

An explicit self-communication channel (e.g., text-like or symbol-like private tokens) can improve performance and sample efficiency on tasks requiring long-horizon credit assignment and planning, beyond what latent recurrence/planning alone achieves. *Selective* self-communication may be an efficient option in a broader action space that also includes direct action and latent deliberation [Nye et al., 2021, Wei et al., 2022, Zelikman et al., 2022, Yao et al., 2023a, Kim et al., 2025, Xie et al., 2025, Wang et al., 2026].

5 Experimental agenda

Here we describe one potential proof-of-concept study for testing our hypotheses, starting with environments where observability, controllability, and cost structure are explicit, and then moving toward richer interactive multimodal settings.

In the first instance we recommend toy, partially observed environments with known latent dynamics and controllable bottlenecks. Gridworld-like domains with sensor coarsening, latency, observation noise, and switchable action-set cardinality are particularly useful because they permit direct intervention on the variables that define our unification and cost metrics [Chevalier-Boisvert et al., 2023]. ARC-AGI-3-style[1] tasks are attractive because they emphasize compositional priors and generalization under sparse data and reward [Chollet, 2019]. This allows for direct testing of interactive perception–action–deliberation with explicit compute and self-communication costs.

A potential extension would introduce a pretrained multimodal model as a perception and world-model backbone, together with a lightweight meta-controller that decides at each step whether to acquire more input, act on the environment, or spend budget on private deliberation and adaptation. A frozen backbone plus lightweight adaptation (e.g., LoRA) is a practical starting point, with richer online learning introduced later [Hu et al., 2022]. Each emitted or consumed token can incur an explicit cost. Observation and action channels can be bottlenecked by cardinality limits, noise, or latency, allowing direct manipulation of the quantities used in the unification proxy. Additional actions aimed at modifying interface bottlenecks can be included to study H2.

6 Related work

Our position is most directly aligned with the intrinsic-motivation and developmental-robotics literature, especially work that treats curiosity as *learning progress* [Schmidhuber, 2010, Oudeyer et al., 2007, Itti and Baldi, 2009, Gottlieb et al., 2013, Pathak et al., 2017, Burda et al., 2019]. A second pillar is empowerment and information-theoretic agency. Empowerment formalizes agent-centric control as a channel-capacity-like quantity from action sequences to future observations and provides a principled bridge between control, exploration, and interface design [Klyubin et al., 2005a,b, Salge et al., 2014, Jung et al., 2011]. We extend this perspective by pairing empowerment plasticity and by embedding both within a resource-cost objective [Massey, 1990, Kramer, 1998]. We also take inspiration from information-theoretic and thermodynamic treatments of learning and decision-making [Still, 2009, Still et al., 2012, Sivak and Crooks, 2012, Ortega and Braun, 2013].

We draw heavily on predictive-processing, active-inference, and world-models, all of which emphasize predictive internal structure and action under uncertainty [Rao and Ballard, 1999, Friston, 2010, Aitchison and Lengyel, 2017, Ha and Schmidhuber, 2018, Hafner et al., 2023]. Most relevant is the idea that good internal models support efficient action and adaptation.

Our discussion of language and self-communication intersects several lines of work: extended cognition and cognitive offloading [Clark and Chalmers, 1998, Kirsh and Maglio, 1994, Risko and Gilbert, 2016]; resource-rational cognition and metareasoning [Russell and Subramanian, 1995, Lieder and Griffiths, 2020, Callaway et al., 2018]; adaptive computation mechanisms [Graves, 2016, Banino et al., 2021, Snell et al., 2025]; and recent reasoning systems that partially decouple internal deliberation from output verbalization [Nye et al., 2021, Wei et al., 2022, Zelikman et al., 2022, Yao et al., 2023a, Kim et al., 2025, Xie et al., 2025, Wang et al., 2026]. We argue that language-like traces are one optional channel in a broader token/action budget.

[1] `https://arcprize.org/arc-agi/3`

Benchmark and systems-level discussions of intelligence emphasize priors, generalization, and real-world constraints [Legg and Hutter, 2007, Hendrycks et al., 2021, Srivastava et al., 2023, Liang et al., 2023, Jimenez et al., 2024]. ARC-AGI-style benchmarks highlight the importance of skill-acquisition efficiency [Chollet, 2019], and discussions of autonomous machine intelligence and human-like learning emphasize the need for richer world models and action-grounded training environments [Lake et al., 2017, LeCun, 2022].

7 Discussion

Limitations. We highlight several limitations and invite feedback from the community and suggestions for improvement. First, non-equivalence between prediction and empowerment is probably common in practice. We expect scenarios in which these quantities diverge because of reward misspecification, hidden confounders, or severe partial observability. Second, some of the most interesting quantities in our position are expensive to estimate and may require loose variational bounds. Finally, human-likeness remains inherently multi-objective. Similarity in constraints may improve interpretability in some settings, but it does not by itself imply safety or desirable behavior across tasks. Our manifold view of intelligence highlights the need for a better science of how specific human-related constraints and frames of reference shape and define human intelligence and the implications on the broader manifold.

Alternative views. A first objection is that better interfaces are not always more lossless or higher bandwidth. More information can increase compute cost and be distracting. For this reason, our notion of interface quality is reward-relative and cost-sensitive. A second objection is that human-like constraints do not imply safe or aligned behavior. We agree. The weaker claim is that constraint mismatch is one source of failure and should be measured and accounted for. A third objection is that self-communication can become verbose, brittle, or uninterpretable. We therefore argue that private reasoning channels should be evaluated under explicit bandwidth and utility costs, and compared against latent deliberation and direct action under matched budgets.

Conclusion. We make the argument for treating AI systems as *environment-embedded, resource-constrained agents*. We formalized this stance with (i) curiosity-as-learning-progress as an intrinsic development signal, (ii) explicit budgets over observation, actuation, compute, and memory in a single objective, and (iii) *unification* as an experimentally manipulable notion of interface quality, including language and self-communication as selective information bottlenecks. We hope our framing helps shift discussion from single-score capability to a more grounded science of agency under constraints, and we invite criticism, counterexamples, and sharper theoretical framing and experimental work.

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